

KAIXIN ZHU

 Google scholar

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 HuggingFace

 github.com/Chriszkxxx

EDUCATION

9/2021 - 6/2025	Hong Kong Baptist University B.S. in Artificial intelligence • First class honors degree	Hong Kong
8/2026 - 8/2027	Nanyang Technological University MSAI	Singapore

RESEARCH AND INTERNSHIP EXPERIENCE

3/2024 - Present	Peking University Research Assistant since March 2024, working with Prof. Wentao Zhang on World Models and multimodal generation, with a focus on 3D and 4D generation. Contributed to the development of the OpenWorldLib unified World Model codebase and the DataFlow platform, helping standardize World Model definitions, training pipelines, and evaluation benchmarks for open-source research. In 4D generation, co-developed Trans4D, a text-to-4D diffusion framework, built WideRange4D, a large-scale benchmark for motion reconstruction, and developed Process4D, a progressive spatiotemporal adaptation method that achieves strong performance on large-motion scenes. In 3D generation, contributed to 3DGen-R1 (CVPR 2026), exploring reinforcement learning for text-to-3D autoregressive models through hierarchical optimization and reward design, and contributed to the development of VGGT-Edit, a feed-forward framework for native 3D scene editing. World Models / Generative AI	Beijing
7/2026 - Present	Unity Led the development of the Unity Avatar Asset Pipeline using Python Editor Scripting to automate the import, validation, and integration of Avatar Mesh, Skeleton, and Animation Sequence assets. Defined Unity-side asset import standards and built automated batch processing based on these standards. To address the lack of skeleton binding information in generative model outputs, developed an automated rigging and skinning pipeline with bind-pose export, enabling generated meshes to be directly driven by skeletons and used in Unity. Further analyzed the differences between the joint structure of SMPL and standard humanoid rigs such as Mixamo in terms of bone topology, coordinate systems, and rotation representations. Designed a cross-skeleton bone mapping strategy and implemented World-space Delta based Motion Retargeting, completing end-to-end deployment and validation of motion assets in Unity. 3D / 4D	Beijing

PROJECTS

OpenWorldLib: A Unified Codebase for Advanced World Models

- [Github](#): 855 stars

WideRange4D: Enabling High-Quality 4D Reconstruction with Wide-Range Movements and Scenes

- [Github](#): 112 stars
- [Huggingface](#): **10k+** Downloads

3DGen-R1: Are We Ready for RL in Text-to-3D Generation? A Progressive Investigation

- [Github](#): 112 stars

Trans4D: Realistic Geometry-Aware Transition for Compositional Text-to-4D Synthesis

- [Github](#): 56 stars

PUBLICATIONS

Are We Ready for RL in Text-to-3D Generation? A Progressive Investigation

Yiwen Tang*, Zoey Guo*, **Kaixin Zhu***, Ray Zhang*, Qizhi Chen, Dongzhi Jiang, Junli Liu, Bohan Zeng, Haoming Song, Delin Qu, Tianyi Bai, Dan Xu, Wentao Zhang, Bin Zhao

Accepted by **CVPR 2026**: Computer Vision and Pattern Recognition 2026

3DGen-R1 is currently accessible on the [arXiv platform](#).

VGGT-Edit: Feed-forward Native 3D Scene Editing with Residual Field Prediction

Kaixin Zhu*, Yiwen Tang*, Yifan Yang*, Renrui Zhang*, Bohan Zeng, Ziyu Guo, Ruichuan An, Zhou Liu, Qizhi Chen, Delin Qu, Jaehong Yoon, Wentao Zhang

VGGT-Edit is currently accessible on the [arXiv platform](#).

WideRange4D: Enabling High-Quality 4D Reconstruction with Wide-Range Movements and Scenes

Ling Yang*, **Kaixin Zhu***, Juanxi Tian*, Bohan Zeng*, Mingbao Lin, Hongjuan Pei, Wentao Zhang, Shuicheng Yan

WideRange4D is currently accessible on the [arXiv platform](#).

Research on World Models Is Not Merely Injecting World Knowledge into Specific Tasks

Bohan Zeng*, **Kaixin Zhu***, Daili Hua*, Bozhou Li*, Chengzhuo Tong*, Yuran Wang*, Xinyi Huang*, Yifan Dai*, Zixiang Zhang*, Yifan Yang*, Zhou Liu, Hao Liang, Xiaochen Ma, Ruichuan An, Tianyi Bai, Hongcheng Gao, Junbo Niu, Yang Shi, Xinlong Chen, Yue Ding, Minglei Shi, Kai Zeng, Yiwen Tang, Yuanxing Zhang, Pengfei Wan, Xintao Wang, Wentao Zhang

This work is currently accessible on the [arXiv platform](#).

OpenWorldLib: A Unified Codebase and Definition of Advanced World Models

DataFlow Team, Bohan Zeng, Daili Hua*, **Kaixin Zhu***, Yifan Dai*, Bozhou Li*, Yuran Wang*, Chengzhuo Tong*, Yifan Yang*, Mingkun Chang*, Jianbin Zhao*, Zhou Liu, Hao Liang, Xiaochen Ma, Ruichuan An, Junbo Niu, Zimo Meng, Tianyi Bai, Meiyi Qiang, Huanyao Zhang, Zhiyou Xiao, Tianyu Guo, Qinhan Yu, Runhao Zhao, Zhengpin Li, Xinyi Huang, Yisheng Pan, Yiwen Tang, Yang Shi, Yue Ding, Xinlong Chen, Hongcheng Gao, Minglei Shi, Jialong Wu, Zekun Wang, Yuanxing Zhang, Xintao Wang, Pengfei Wan, Yiren Song, Mike Zheng Shou, Wentao Zhang

OpenWorldLib is currently accessible on the [arXiv platform](#).

Trans4D: Realistic Geometry-Aware Transition for Compositional Text-to-4D Synthesis

Bohan Zeng*, Ling Yang*, Siyu Li, Jiaming Liu, Zixiang Zhang, Juanxi Tian, **Kaixin Zhu**, Yongzhen Guo, Fu-Yun Wang, Minkai Xu, Stefano Ermon, Wentao Zhang

Trans4D is currently accessible on the [arXiv platform](#).

ACHIEVEMENTS

2025	First Class Scholarship	Campus
2024	Second Class Scholarship	Campus
2024-2023	Best Poster Award of Computer Science and Technology	Campus

SERVICES

2025	Presenter Introducing our work WideRange4D at the TechBeat
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MEMBERSHIP

2024	Student Membership Institute of Electrical and Electronics Engineers (IEEE)
2024	Student Membership China Computer Federation (CCF)

SKILLS & INTERESTS

Programming:	Python, R, Java, HTML, SQL.
Miscellaneous:	Linux, LATEX, Microsoft Office, Git.
Interests:	Reading, Photography, Travel